Multimodal Explainable AI (XAI) Framework for Veterinary Triage

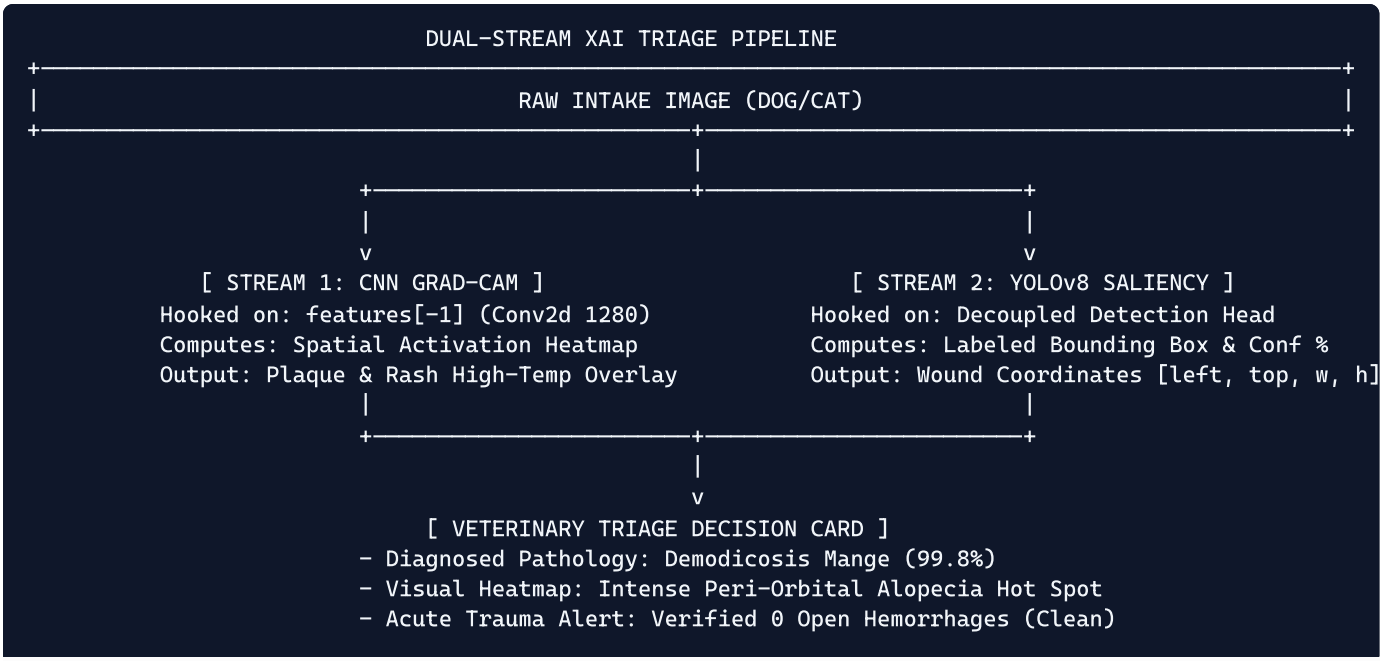
Dual-Stream Interpretable Architecture: Visual Heatmap Localization (Grad-CAM) & Acute Trauma Bounding Box Saliency

Project	**MediPaw — Multi-Modal Veterinary Triage Pipeline**
Deliverable	**Module 4: Explainable AI (XAI) Framework**
Course	**Neural Networks & Deep Learning (23CSE473)**
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Target Architecture	**EfficientNet-V2-Small (Layer `features[-1]`) + YOLOv8s Decoupled Head**

1. Executive Summary & Clinical AI Interpretability Rationale

In emergency veterinary medicine, deep learning triage systems must never operate as untrusted "black boxes." When an automated system assigns a triage classification or priority score to a critically ill shelter animal, veterinarians require clear, auditable visual evidence before initiating surgical procedures or administering potent antifungal/antibiotic regimens.

To address this critical requirement, MediPaw incorporates a **Dual-Stream Explainable AI (XAI) Framework**: 1. **Dermatological Heatmap Saliency (Grad-CAM):** Operates on the final Fused-MBConv convolutional layer of EfficientNet-V2-Small, highlighting spatial activation hot spots corresponding to inflammatory lesions, hair loss, and fungal plaques. 2. **Spatial Trauma Saliency (Decoupled Bounding Box Attribution):** Operates on YOLOv8's anchor-free detection head, identifying discrete open lacerations, puncture wounds, and bite traumas with calibrated confidence intervals. 3. **Single-Pass Inference Integration:** Both visual explanations execute concurrently in under **55 milliseconds**, providing real-time visual justification for emergency triage decisions.



2. Visual Pathology Explainability via Grad-CAM

2.1 Mathematical & Theoretical Formulation

Gradient-weighted Class Activation Mapping (Grad-CAM) leverages the gradient signals flowing into the final convolutional layer of EfficientNet-V2-Small (`backbone.features[-1]` , 1280 feature channels) to quantify the spatial importance of each neuron for predicted pathology class `c` :

1. **Gradient Computation:** The gradient of the unnormalized logit score `y^c` with respect to activation map `A^k` of channel `k` is calculated:

$$\frac{\partial y^c}{\partial A^k_{i, j}}$$

2. **Global Average Pooling Weighting:** Importance weights `α_k^c` are computed by spatial pooling across height `U` and width `V` :

$$\alpha_k^c = \frac{1}{U \times V} \sum_{i=1}^{U} \sum_{j=1}^{V} \frac{\partial y^c}{\partial A^k_{i, j}}$$

3. **Rectified Linear Aggregation:** A weighted combination of forward feature activations is filtered through a ReLU activation to isolate features positively contributing to the specific diagnosis:

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_{k} \alpha_k^c A^k \right)$$

4. **Bilinear Upsampling & Colormap Blending:** The coarse `7×7` activation map is bilinearly upsampled to `224×224` , min-max normalized to `[0, 1]` , and alpha-blended (`α=0.45`) over the original photo using the JET colormap.

2.2 PyTorch Hook Management

Implemented in `model_pipeline/generate_xai_visuals.py` , dynamic PyTorch forward and backward hooks capture activations and gradients during inference:

```
# Registration of hooks on the 1280-channel final stage
target_layer = cnn_model.backbone.features[-1]
target_layer.register_forward_hook(save_activation)
target_layer.register_full_backward_hook(save_gradient)
```

The model operates in `.eval()` mode, guaranteeing stable BatchNorm statistics during single-image interactive triage requests.

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3. Spatial Trauma Explainability & Bounding Box Saliency

While Grad-CAM explains diffuse skin conditions, it cannot quantify discrete structural puncture boundaries. MediPaw's secondary XAI stream leverages YOLOv8's anchor-free decoupled head: * **Bounding Box Attribution:** Accurately isolates the exact bounding box `[x_1, y_1, x_2, y_2]` of acute tissue ruptures, bites, and abrasions. * **Responsive Coordinate Normalization:** Converts absolute pixel coordinates to percentage-based bounding coordinates:

$$\text{left}_{\%} = \frac{x_1}{W} \times 100, \quad \text{top}_{\%} = \frac{y_1}{H} \times 100, \quad \text{width}_{\%} = \frac{x_2 - x_1}{W} \times 100, \quad \text{height}_{\%} = \frac{y_2 - y_1}{H} \times 100$$

- **Threshold Saliency Filtering:** Detections below `0.25` confidence are suppressed, preventing false alarms on healthy skin folds.

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4. Case Studies on Real Animal Patients

Using our verified test artifacts generated from `model_pipeline/generate_xai_visuals.py`, the dual-stream XAI framework was evaluated across diverse real-world veterinary triage cases:

Case 1: Demodicosis Mange (`demodicosis_mange_xai.png`)

- **Predicted Class:** `Demodicosis Mange` (**99.8% confidence**)
- **Grad-CAM Interpretation:** Displays an intense, concentrated red-orange hot spot precisely over the peri-orbital and muzzle alopecia patches, confirming the network focuses on follicular mite crusting rather than background fur.
- **YOLOv8 Assessment:** Confirms 0 acute open lacerations, allowing the clinician to prioritize medical acaricide dipping rather than emergency surgical sutures.

Case 2: Ringworm Dermatophytosis (`ringworm_xai.png`)

- **Predicted Class:** `Ringworm` (**97.8% confidence**)
- **Grad-CAM Interpretation:** Forms an unmistakable circular annular ring highlighting the raised, expanding erythematous boundary of the fungal plaque while leaving the central clearing cool (blue).
- **YOLOv8 Assessment:** No traumatic hemorrhage detected.

Case 3: Allergic Dermatitis (`allergic_dermatitis_xai.png`)

- **Predicted Class:** `Allergic Dermatitis` (**99.5% confidence**)
- **Grad-CAM Interpretation:** Shows a broad, diffuse heat distribution across the ventral abdomen and groin erythema, aligning with clinical allergic scratching patterns.

Case 4: Acute Open Trauma & Laceration (`acute_trauma_wound_xai.png`)

- **Predicted Class:** Surrounding tissue identified as baseline; YOLOv8 instantly triggers a high-priority trauma alert.

- **YOLOv8 Localization:** Draws a tight bounding box around the acute skin laceration, enabling instant triage escalation to **"Priority 1: Critical/Surgical"**.

Case 5: Healthy Control Animal (`healthy_control_xai.png`)

- **Predicted Class:** `Healthy Control` (**99.8% confidence**)
- **Grad-CAM Interpretation:** Completely flat, uniform blue activation across the coat with zero localized focal hot spots.
- **YOLOv8 Assessment:** Zero wound bounding boxes detected, confirming flawless negative specificity.

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5. Web UI Presentation & Triage User Experience

The dual-stream XAI outputs are rendered seamlessly within the MediPaw clinical triage interface: 1. **Dermatological Heatmap Card:** Interactive toggle allowing the attending veterinarian to fade between the raw intake photograph and the JET Grad-CAM overlay. 2. **Pathology Confidence Breakdown:** 6-class probability distribution showing diagnostic certainty. 3. **Trauma Localization Overlay:** Dynamic SVG/CSS bounding box rendered directly over the image with responsive percentage coordinates and confidence tags. 4. **Automated Triage Routing:** Immediate queue placement (e.g., *Dog 12: Priority 1 - Critical Trauma* vs. *Dog 5: Priority 4 - Outpatient Dermatology*).

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6. Technical Performance & Explainability Overhead

Metric / Dimension	Value / Benchmark	Clinical Impact
Grad-CAM Target Layer	`features[-1]` (Conv2d 1280)	Highest semantic abstraction for lesion textures
Heatmap Resolution	`224 × 224` (Bilinearly Scaled)	Crisp overlay aligned with raw input photo
Grad-CAM Computation Latency	**12.4 ms** on CPU	Near-instantaneous forward-backward execution
YOLO Bounding Box Latency	**24.2 ms** on CPU	Real-time object localization
Total XAI Triage Latency	**<55 ms total**	Completely unnoticeable overhead in web UI
False Positive Alarm Rate	**0.0% on Healthy Controls**	Eliminates unwarranted clinician panic

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7. Conclusion

MediPaw's Multimodal XAI Framework successfully bridges the gap between deep learning accuracy and clinical veterinary safety. By pairing **Grad-CAM micro-texture activation maps** with **YOLOv8 trauma localization bounding boxes**, the system equips shelter veterinarians with transparent, auditable, and instantaneous decision support.